Laguerre-Gaussian (l=0)

Maximum intensity

Laguerre-Gaussian equation (for p=0, i.e. LG0l), with cylindrical coordinates of radial $r \geq 0$, azimuthal $\varphi \in [0, 2\pi)$ and $z = 0$, reduces to

$$E_{\mathrm{LG}0l} = E_0 \exp(il\varphi) \left(\frac{\sqrt{2}r}{\omega_0} \right)^{|l|} \exp\left(-\frac{r^2}{\omega_0^2} \right)$$

for the electric field. Then the intensity corresponds to

$$I_{\mathrm{LG}0l} \propto \langle E_{\mathrm{LG}0l} | E_{\mathrm{LG}0l} \rangle = |E_0|^2 \left(\frac{2r^2}{\omega_0^2} \right)^{|l|} \exp\left(-\frac{2r^2}{\omega_0^2} \right)$$

with the maximum intensity at r where

$$\frac{d}{dr} I_{\mathrm{LG}0l} \propto \frac{2|l|}{r} - \frac{4r}{\omega_0^2} = 0$$

or in other words,

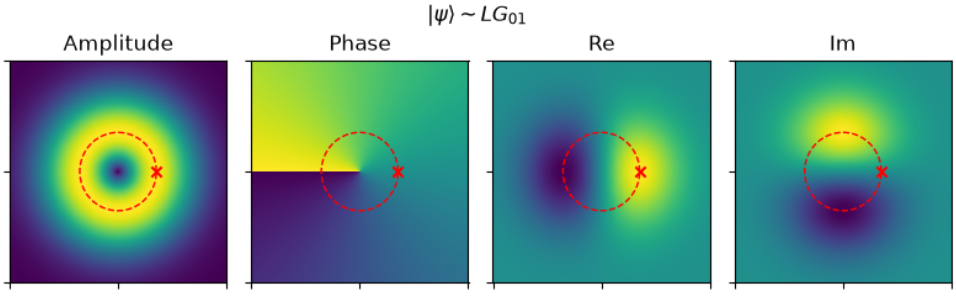
$$r = \omega_0 \sqrt{\frac{|l|}{2}}$$

Note that the LG basis is defined by the characteristic beam waist.

Phase dislocation

For a Gaussian source amplitude, generating the LG0l mode simply involves imparting an additional phase, given by:

$$\phi = \mathrm{Arg}[E_{\mathrm{LG}0l}] = l\varphi$$



In the case of a coherent superposition of LG00 and LG0l (and with a phase term of $-\exp(-i\delta)$ without loss in generality), evaluating the sum of the electric fields yields

$$\begin{aligned} E &= \tfrac{1}{\sqrt{2}}(E_{\mathrm{LG}00} - e^{-i\delta} E_{\mathrm{LG}0l}) \\ &= \frac{1}{\sqrt{2}} E_0 \exp\left(-\frac{r^2}{\omega_0^2} \right) \left[1 - e^{i(l\varphi - \delta)} \left(\frac{\sqrt{2}r}{\omega_0} \right)^{|l|} \right] \end{aligned}$$

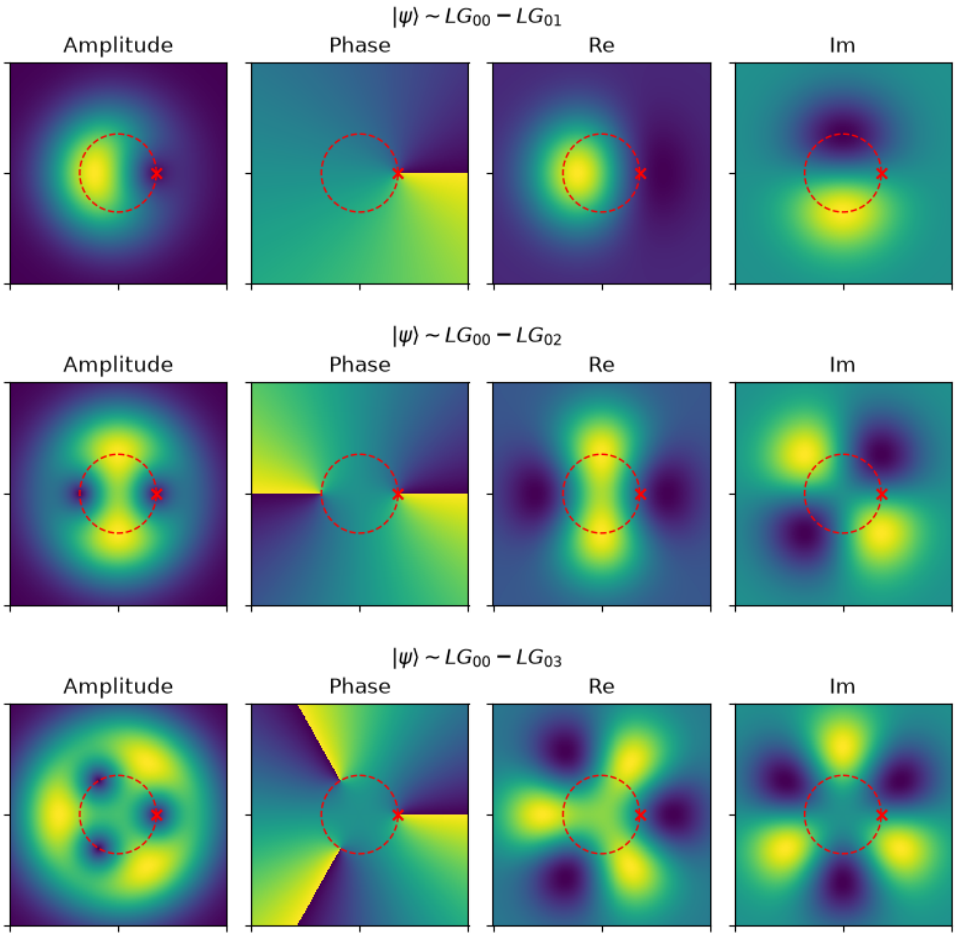
and the phase correspondingly yields

$$\phi = \mathrm{Arg}[E] = \tan^{-1}\left(\frac{C \sin(l\varphi - \delta)}{1 - C\cos(l\varphi - \delta)} \right), \qquad C \equiv \left(\frac{\sqrt{2}r}{\omega_0} \right)^{|l|}$$

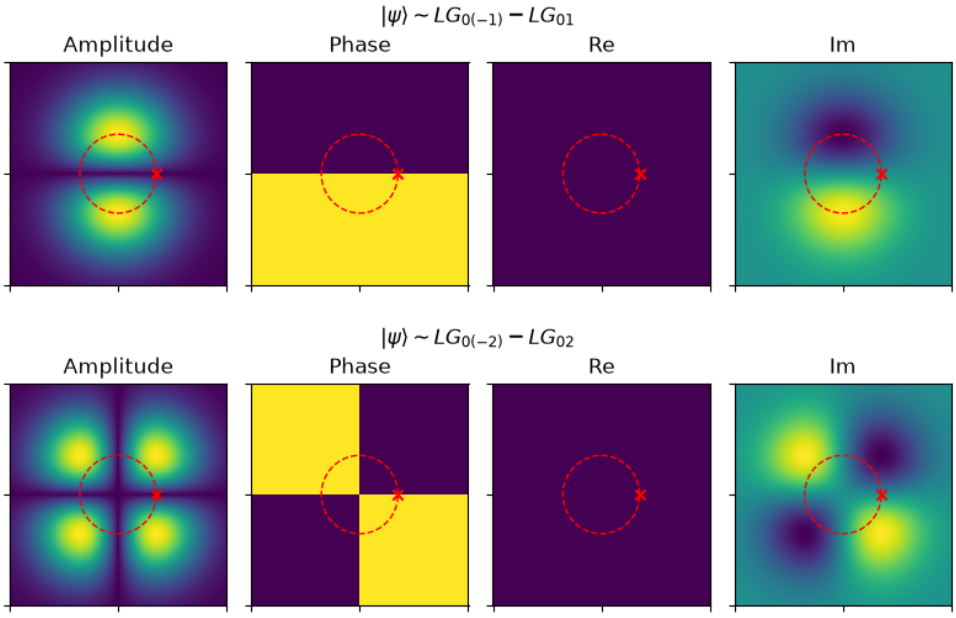
The dislocations are therefore located at the poles where $C\cos(l\varphi-\delta)=1$, leading to the solutions

$$r=w_0/\sqrt{2},\qquad \varphi=(2k\pi+\delta)/l,\qquad k\in\mathbb{Z}$$

Visualizations below are marked with a circle at $r=w_0/\sqrt{2}$, as well as the phase dislocation along $\varphi=0$ given a relative phase $\delta=0\pi$:

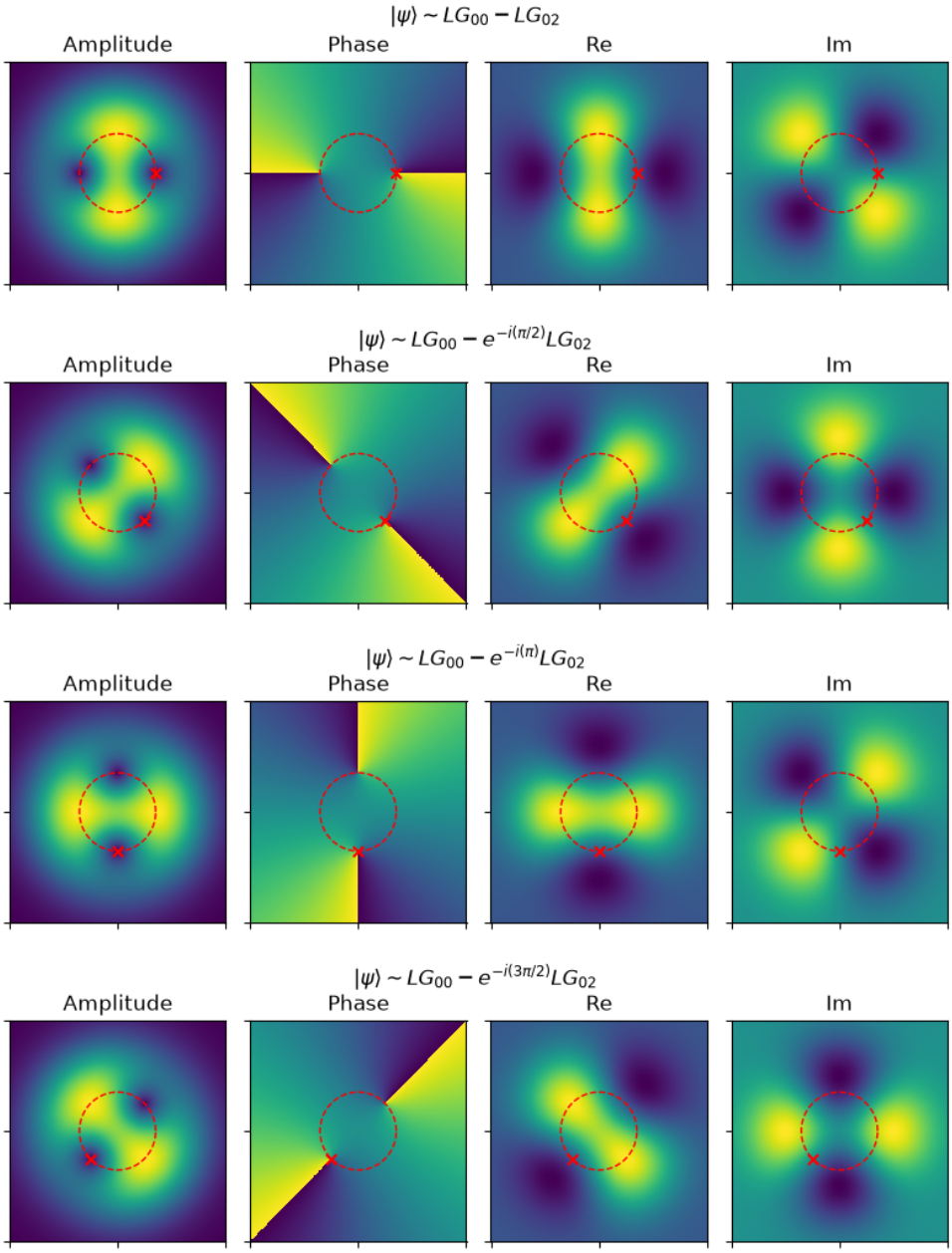


This was first experimentally observed by Zeilinger's group in (Mair, 2001). Similar solutions between different LG terms can be derived as well, with the solution for $LG_{0l}+LG_{0(-l)}$ generating flat phase profiles, as depicted also in (Jack, 2009):



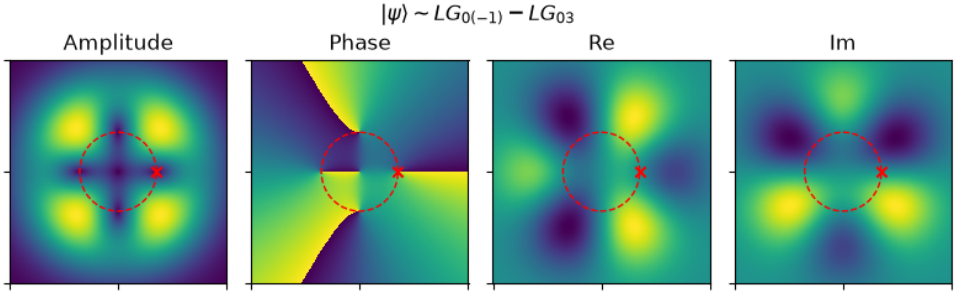
Mutually unbiased basis (MUB)

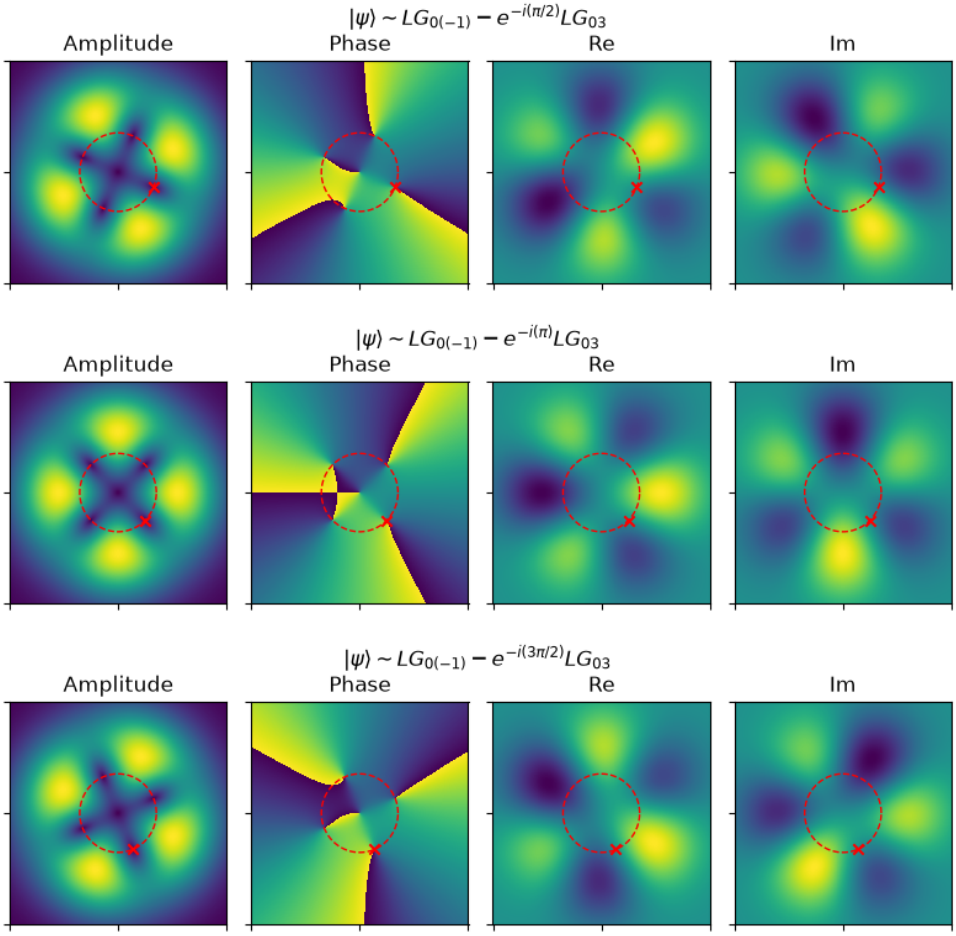
This is the term commonly used in the holography field, which corresponds to the set of complementary bases. Cycling through the relative phase in $[0, 2\pi)$ yields a circular rotation, in the case of a rotationally symmetric phase map. Depicted for $LG_{00} + \exp(i\delta) LG_{02}$ below:



Note that the zeros of the coherent sum also correspond exactly to the locations of the phase dislocations.

In the case of non-rotationally symmetric phase maps, there exists multiple such zeros, generally (1) on the circle $r = w_0/\sqrt{2}$ (with the number of roots corresponding to the difference in topological orders), and (2) one at the center. The corresponding phase evolution is also not directly obvious, here depicted for $LG_{0(-1)} + \exp(i\delta) LG_{03}$ (noting the $|3 - (-1)| = 4$ roots and central zero):





References

- (Mair, 2001): A. Mair, et. al., “Entanglement of the orbital angular momentum states of photons,” Nature **412** 313-316 (2001)
- (Jack, 2009): B. Jack, et. al., “Precise quantum tomography of photon pairs with entangled orbital angular momentum,” New J. Phys. **11** 103024 (2009)